

TermScope: AI- Powered Legal Document Risk Analyzer

A PROJECT REPORT

Submitted by

BANTY KUMAR (24MIM10234)

MUSKAN JAIN (24MIM10164)

AYUSH KUMAR (24MIM10153)

YASH PATWA (24MIM10112)

SOUMYA PANDEY (24MIM10232)

INTEGRATED MASTER OF TECHNOLOGY

In

ARTIFICIAL INTELLIGENCE

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING



VIT BHOPAL UNIVERSITY

KOTRIKALAN, SEHORE

MADHYA PRADESH-466114

SEPTEMBER 2025

BONAFIDE CERTIFICATE

This is to Certified that this project report titled “ **TermScope: AI- Powered Legal Document Risk Analyzer**” is the bonafide work of “Banty Kumar (24MIM10234), Muskan Jain (24MIM10164), Soumya Pandey (24MIM10234), Yash Patwa (24MIM10112), Ayush Kumar (24MIM10153) ” who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

PROGRAM CHAIR

Name: Dr S. Devaraju

Designation: Senior Assistant Professor

Date:

REVIEWER 1

Name: Dr.Vinesh Kumar

Designation: Assistant Professor

Date:

SUPERVISOR

Name: Dr. J Subash Chandra Bose

Designation: Senior Associate Professor

Date:

REVIEWER 2

Name: Dr. Pradeep Kumar

Designation: Assistant Professor

Date:

The Project exhibition I Examination is held on _____

ACKNOWLEDGEMENT

I am deeply indebted to **Dr. G. Viswanathan**, Chancellor of VIT Bhopal University, and **Dr. T.B. Sridharan**, Pro Vice-Chancellor, VIT Bhopal University, for their unwavering support and guidance. Their leadership and vision have been a constant source of inspiration throughout my academic journey.

Express my profound gratitude to **Dr. Pon Harshavardhan**, Dean of SCAI, for his encouragement and invaluable support at every stage, which significantly contributed to the successful completion of my project and degree.

My heartfelt thanks go to **Dr. S. Devaraju**, Program Chair of Integrated M.Tech- Artificial Intelligence, and **Dr. Venkat Prasad Padhy**, Division Head of the AI Division, for their timely assistance, guidance, and invaluable insights during the course of my research. A special note of appreciation reserved for my supervisor, **Dr. J Subash Chandra Bose**, whose constant support, patience, and expertise were instrumental in completing this project. His guidance, encouragement, and clarity in addressing my queries made this journey both meaningful and enriching.

I would like to express my sincere gratitude to **Dr. Vinesh Kumar**, and **Dr. Pradeep Kumar**, for their valuable feedback and insightful reviews. Lastly, I extend my deepest gratitude to my parents and friends, whose unwavering encouragement and support have been my driving force throughout this research endeavor. Their belief in me has been a source of strength and motivation.

LIST OF SYMBOLS, NOMENCLATURE, AND ABBREVIATIONS USED

ABBREVIATIONS AND SYMBOLS:

AI	- Artificial Intelligence
API	- Application Programming Interface
CCPA	- California Consumer Privacy Act
CPU	- Central Processing Unit
CSS	- Cascading Style Sheets
DOM	- Document Object Model
GDPR	- General Data Protection Regulation
GPT	- Generative Pre-trained Transformer
HTML	- Hyper Text Markup Language
HTTP	- Hyper Text Transfer Protocol
JSON	- JavaScript Object Notation
JS	- JavaScript
ML	- Machine Learning
NLP	- Natural Language Processing
OCR	- Optical Character Recognition
PDF	- Portable Document Format
RAM	- Random Access Memory
SUS	- System Usability Scale
T&C	- Terms and Conditions
UI	- User Interface
UX	- User Experience

TECHNICAL TERMS

Chrome Extension - A software program that extends the functionality of Google Chrome

Manifest V3 - The third version of Chrome extension manifest specification

Content Script - JavaScript that runs in the context of web pages

Background Script - Extension service worker that handles events

Popup - The small window that appears when clicking the extension icon

DOM Extraction - Process of retrieving content from web page Document Object Model

Heuristic Analysis - Rule-based pattern matching approach

Pattern Matching - Technique for finding specific text patterns using regular expressions

API Integration - Connecting to external services through Application Programming Interfaces

SYMBOLS AND NOTATIONS

→ - Data flow/Process direction

% - Percentage

ms - Milliseconds

s - Seconds

< - Less than

> - Greater than

± - Plus/minus (statistical variation)

LIST OF FIGURES AND GRAPHS

FIGURE NO.	TITLE	PAGE NO.
4.1	Component Diagram of TermScope Chrome Extension	10
4.2	Chrome Extension Interface Design	11
5.1	Chrome Extension Content Analysis Workflow	12

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
2.1	Comparison of Existing T&C Analysis Tools	06
3.1	Hardware and Software Requirements	07
3.2	Functional Requirements Analysis	08
5.1	Performance Results Table	15
6.1	Application in Different Domains	20

ABSTRACT

This project aims to address the critical challenge that 97% of users accept Terms and Conditions without reading them due to excessive length, complexity, and legal jargon. The primary objective is to develop an AI-powered Chrome extension that automatically analyzes T&C documents to protect users from hidden risks, unfair clauses, and compliance violations, thereby democratizing access to legal document understanding. The project implements a hybrid analysis approach combining local heuristic pattern matching with optional OpenAI GPT-4 integration. The Chrome extension (Manifest V3) utilizes advanced DOM content extraction to retrieve text from web pages, followed by a dual-analysis pipeline: (1) Local rule-based analyzer using regular expressions for risk detection, and (2) Cloud-based AI analysis for comprehensive understanding when enabled.

The system architecture employs modern web technologies including JavaScript, Chrome Extension APIs, and RESTful API integration, with a privacy-first design that processes sensitive data locally by default. The implemented TermScope extension successfully processes T&C documents with 78% accuracy using local analysis and 92% accuracy with AI integration, reducing analysis time from an average of 42 minutes to under 5 seconds. User testing with 25 participants demonstrated 94% task completion rate, 88% comprehension improvement, and 4.3/5.0 satisfaction score. The project validates that automated legal document analysis can significantly enhance consumer protection while maintaining privacy standards. The hybrid approach proves effective in balancing performance, accuracy, and user control, providing a scalable foundation for future legal technology applications.

KEYWORDS: Chrome Extension, Terms and Conditions, AI Analysis, Legal Technology, Consumer Protection, Natural Language Processing, Privacy by Design, Automated Document Analysis

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
1	CHAPTER 1: PROJECT DESCRIPTION AND OUTLINE 1.1 Introduction 1.2 Motivation for the Work 1.3 Project Overview with Techniques 1.4 Problem Statement 1.5 Objectives of the Work 1.6 Organization of the Project 1.7 Summary	 01 03 03 03 03 04 04
2	CHAPTER 2: RELATED WORK INVESTIGATION 2.1 Introduction 2.2 Core Area: Legal Document Analysis using NLP 2.3 Existing Approaches/Methods 2.4 Comparative Analysis 2.5 Research Gaps Identified 2.6 Summary	 05 05 05 06 06 06
3	CHAPTER 3: REQUIREMENT ARTIFACTS 3.1 Introduction 3.2 Hardware and Software Requirements 3.3 Specific Project Requirements 3.4 Summary	 07 07 08 09

4	CHAPTER 4: DESIGN METHODOLOGY AND NOVELTY 4.1 Methodology and Goals 4.2 Functional Modules Design 4.3 Software Architecture 4.4 Novelty Aspects 4.5 User Interface Design 4.6 Summary	10 10 10 10 11 11
5	CHAPTER 5: TECHNICAL IMPLEMENTATION & ANALYSIS 5.1 Outline 5.2 System Architecture Diagrams 5.3 Performance Analysis 5.4 Testing Strategy 5.5 Validation Results 5.6 Summary	12 12 13 15 16 17
6	CHAPTER 6: PROJECT OUTCOME AND APPLICABILITY 6.1 Outline 6.2 Key System Features 6.3 Significant Outcomes 6.4 Real-World Applications 6.5 Social Impact 6.6 Inference	18 18 18 19 19 20

7	CHAPTER 7: CONCLUSIONS AND RECOMMENDATIONS 7.1 Outline 7.2 Limitations and Constraints 7.3 Future Enhancements 7.4 Conclusions 7.5 Recommendations	21 21 21 22 22
8	CHAPTER 8: LITERATURE REVIEW 8.1 Introduction 8.2 Core Area: Legal Document Analysis using NLP 8.3 Research Gaps Identified and TermScope's Contribution	23 23 24
	REFERENCES	25